WEEK -3

Minor Project (BTCS607N)-Weekly Progress Report

Project Title: Drone-Based Intelligent System for Apple Orchard Management in Himachal Pradesh.

Group ID: G9

Branch & Section: BTech CSE (AIML with Microsoft), AIDS -A

Date: 04-02-2026

1. Executive Summary

A. **Current Sprint:** Sprint 1 (Issue – 3)

B. **Sprint / Issue Goal:** To analyze the problem domain and design a clear class model showing entities, attributes, and relationships.

C. **Status:** On Track

2. UML & OOAD Progress

A. **Diagrams Completed this week:**

1. Domain Model Diagram
2. Class Diagram (Core Entities, Attributes, Responsibilities, and Relationships)

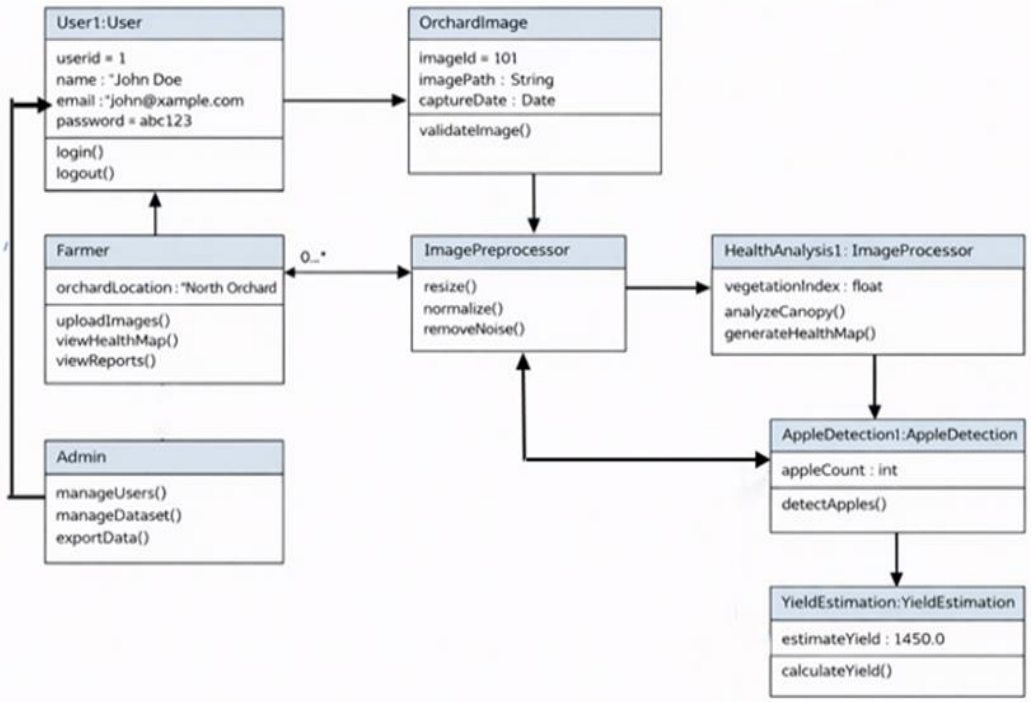
B. **Diagrams Revised:**

1. Improved class diagram structure after domain analysis discussion
2. Added inheritance hierarchy (User → Farmer, Admin)
3. Added processing pipeline classes (ImagePreprocessor, HealthAnalysis, AppleDetection, YieldEstimation)
4. Corrected associations and multiplicities between Farmer, OrchardImage, and Analysis modules
5. Use Case Diagram updated.
6. Actors and system interactions refined

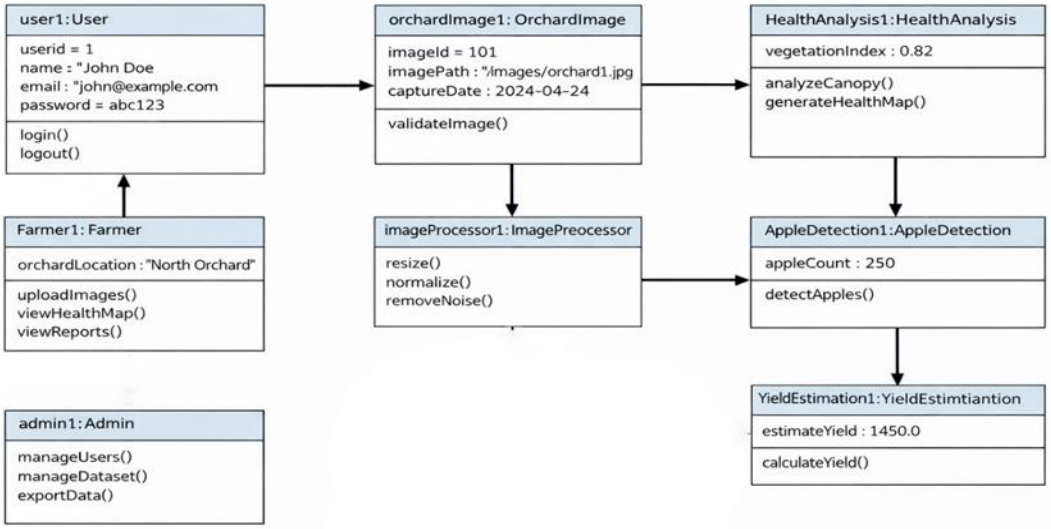
C. **OO Concept Applied:**

1. **Inheritance:** Farmer and Admin derived from User.
2. **Association:** Farmer uploads Orchard Image; modules linked in analysis pipeline.
3. **Encapsulation:** Each class assigned clear responsibilities and operations.
4. **Separation of Concerns:** Detection, preprocessing, and yield estimation handled by separate class.

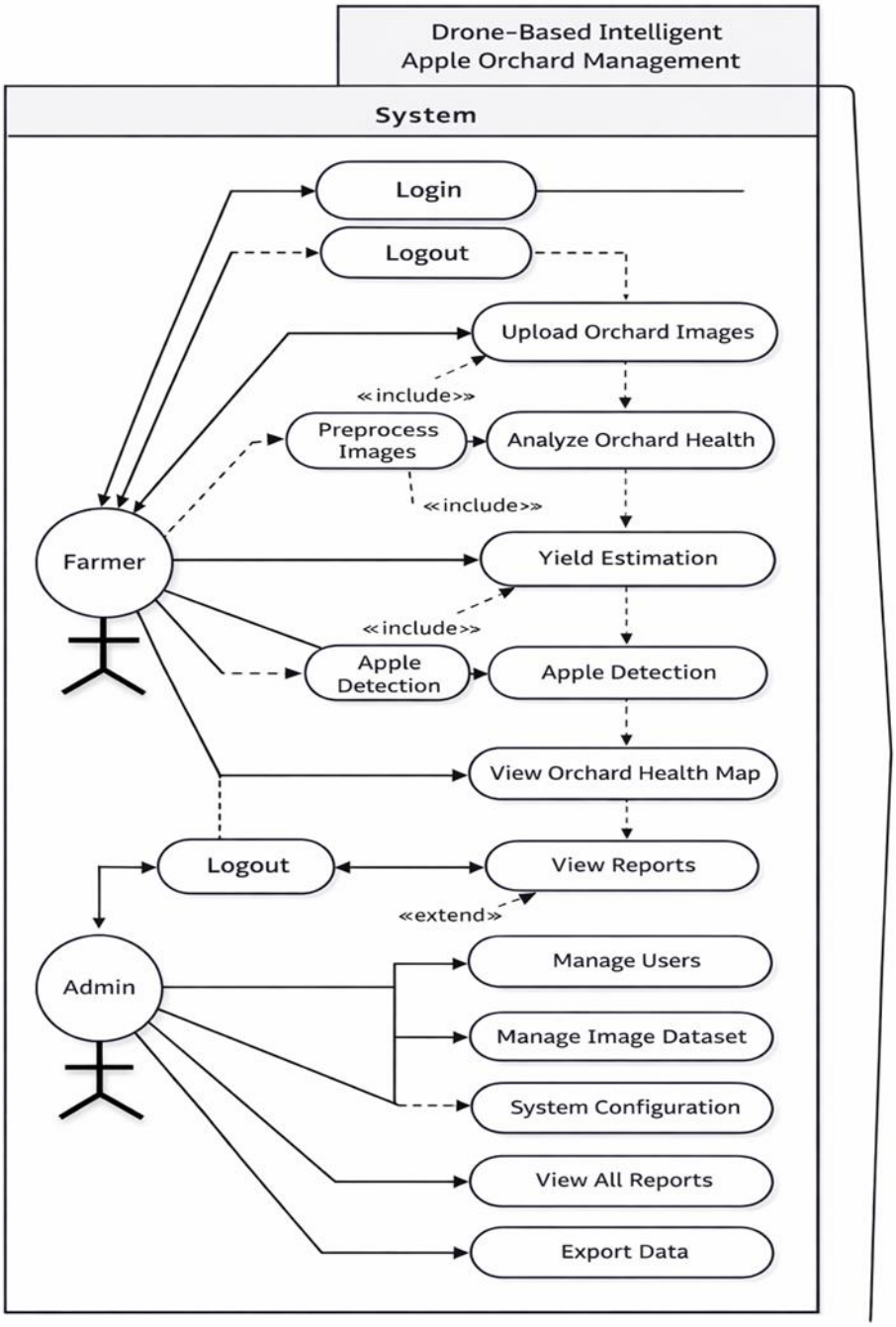
UML Class Diagram – Drone-Based Orchard Management System



UML Object Diagram – Drone-Based Orchard Management System



Updated UML Use-Case Diagram



3. Technical Execution (DevOps & Git)

A. **Jira Tasks Completed:**

1. Identify domain entities
2. Create class diagram (initial)
3. Define attributes & relationships
4. Review class model
5. Use Case Diagram reviewed and updated

Link://ayushshrivastava.atlassian.net/jira/software/projects/DBISFAOM/boards/2?atlOrigin=eyJpIjoiY2I0Y2FjMmQ2MTliNDBlM2FjYzEyN2U1MjdkY2QzNGMiLCJwIjoiaiJ9

B. **Git Activity:**

1. Active Branches: main
2. Merged Pull Requests: Initial documentation setup

Link : https://github.com/yp1818/Drone-Based-Apple-Orchard-Management

C. **CI/CD Status:** Not applicable (design phase – no code yet)

4. Testing Summary

A. **Total Unit Tests:** Not applicable in design phase

B. **Pass Rate:** Not applicable in design phase

C. **Key Bug Identified:** No major issues were identified during this phase.

5. Goals for Next Week

A. Perform dynamic analysis to understand how the system reacts to events and user interactions.

B. Identify and document the main workflow of image capture, processing, analysis, and yield estimation.

C. Design a UML Activity Diagram to represent overall business logic and parallel processing flows.

Mentor Signature **Project Guide Signature**